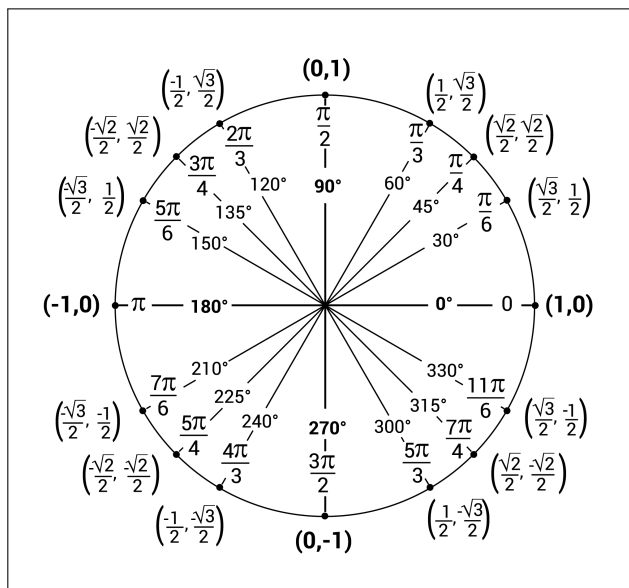


Extra Unit B practice 1

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the final practice worksheets, and the practice problems. The formulas below will be provided on Opportunity B.

Formulas

- $L(x) = f'(a)(x - a) + f(a)$
- MVT: $\frac{f(b) - f(a)}{b - a} = f'(c)$
- Basic derivatives
 - $\frac{d}{dx}(x^n) = nx^{n-1}$
 - $\frac{d}{dx}(e^x) = e^x$
 - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
 - $\frac{d}{dx}(\sin x) = \cos x$
 - $\frac{d}{dx}(\cos x) = -\sin x$
 - $\frac{d}{dx}(\tan x) = \sec^2 x$
 - $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
 - $\frac{d}{dx}(\cot x) = -\csc^2 x$
 - $\frac{d}{dx}(\csc x) = -\csc x \cdot \cot x$
- Derivative rules
 - $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
 - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{[g(x)]^2}$
 - $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$



B1 - Extrema of functions

(a) The graph of a derivative $f'(x)$ is shown on the right.

- i. Find the x -coordinates of all critical points of $f(x)$.

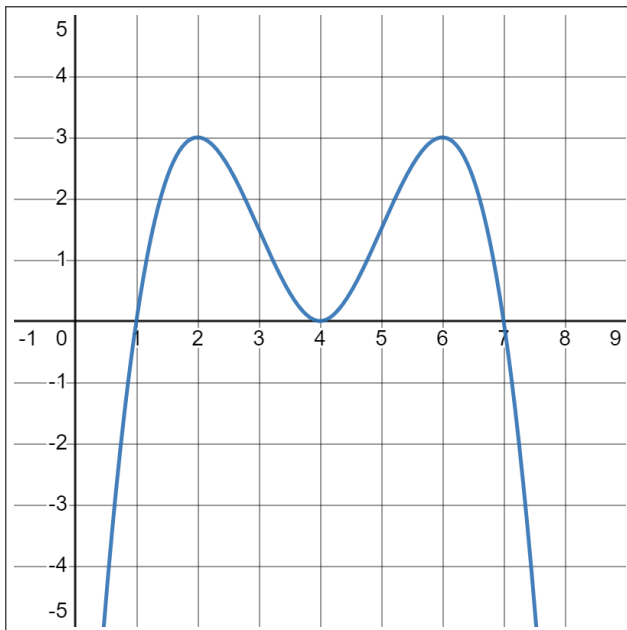
$$x=1, 4, 7$$

- ii. Classify each critical point as a local maximum, a local minimum, or neither.

1: local min

4: neither

7: local max



(b) A function $g(x)$ and its derivative are given below.

$$g(x) = \frac{2x+1}{x^2+2}$$

$$g'(x) = \frac{-2(x+2)(x-1)}{(x^2+2)^2}$$

Find the absolute extreme values of $g(x)$ on the interval $[-1, 4]$. Clearly show each step of your work.

$$g'(x) = 0 \rightarrow x = \underbrace{-2}_{\text{not in } [-1, 4]} \text{ and } x = 1$$

$$g(-1) = \frac{2(-1)+1}{(-1)^2+2} = \frac{-1}{3} \leftarrow \text{abs min}$$

$$g(1) = \frac{2(1)+1}{1^2+2} = \frac{3}{3} = 1 \leftarrow \text{abs max}$$

$$g(4) = \frac{2(4)+1}{4^2+2} = \frac{9}{18} = \frac{1}{2}$$

B2 - The Mean Value Theorem

- (a) Let $f(x) = (x-3)^{-2}$. Show that there is no value of c in $(2, 4)$ such that

$$\frac{f(4) - f(2)}{4 - 2} = f'(c)$$

Then, explain why this does not contradict the Mean Value Theorem.

$$\frac{f(4) - f(2)}{4 - 2} = \frac{(4-3)^{-2} - (2-3)^{-2}}{4 - 2} = 0$$

$$f'(x) = -2(x-3)^{-3} = 0$$

$$\frac{-2}{(x-3)^3} = 0 \quad \text{has no solutions (numerator is never 0).}$$

This doesn't contradict MVT because $f(x)$ is not continuous on $[2, 4]$ (discontinuity @ $x=3$).

- (b) Show that the equation $x^3 - 15x + c = 0$ has at most one root in the interval $[-2, 2]$. Be sure to justify why you can use any theorems you use.

$f(x)$ is differentiable, so we can apply Rolle's Theorem.

$$f'(x) = 3x^2 - 15 = 0 \rightarrow x = \pm\sqrt{5}, \text{ not in } [-2, 2].$$

So $f'(x)$ has no roots in $[-2, 2]$, so by Rolle's Theorem

$f(x)$ can have at most 1 root in $[-2, 2]$.

B3 - Curve sketching

Consider the function $f(x) = e^{-2x}x^4$. The first and second derivatives of this function are given below.

$$f'(x) = -2e^{-2x}x^3(x-2)$$

$$f''(x) = 4e^{-2x}x^2(x-1)(x-3)$$

- (a) Find the x -coordinates of any local maxima and local minima of $f(x)$. Be sure to show your work for full credit.

$$f'(x) = 0 \text{ @ } x = 0, 2$$

$x = 0$ local min

$x = 2$ local max



- (b) Find the x -coordinates of any inflection points of $f(x)$. Be sure to show your work for full credit.

$$f''(x) = 0 \text{ @ } x = 0, 1, 3$$

$x = 1, 3$ are points of inflection



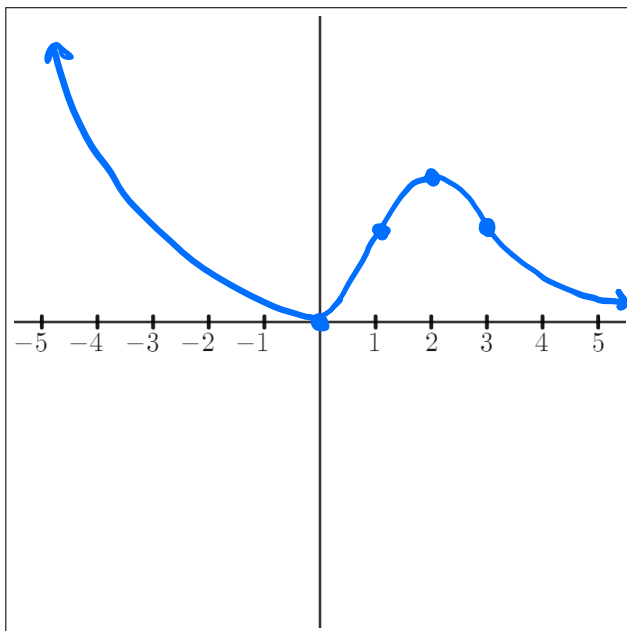
- (c) Find $\lim_{x \rightarrow -\infty} f(x)$.

$$\lim_{x \rightarrow -\infty} e^{-2x} \cdot x^4 \approx e^{\infty} \cdot (-\infty)^4 = \infty$$

- (d) It can be shown that

$$\lim_{x \rightarrow \infty} f(x) = 0 \text{ and } f(0) = 0$$

Use these facts, along with the information you found in parts (a)-(c) to sketch a graph of $f(x)$ on the axes provided to the right. Notice that there is no scale on the vertical axis, so your graph will not be graded by its vertical scale, but all features from parts (a)-(c) should be clearly indicated on your sketch.



B4 - Optimization

- (a) What is the minimum vertical distance between the line $y = 4x + 5$ and the parabola $y = x^2 + 2x + 9$? Be sure to demonstrate, using an appropriate test, that the distance you find is a minimum.

$$\text{vertical dist: } D(x) = (x^2 + 2x + 9) - (4x + 5) = x^2 - 2x + 4$$

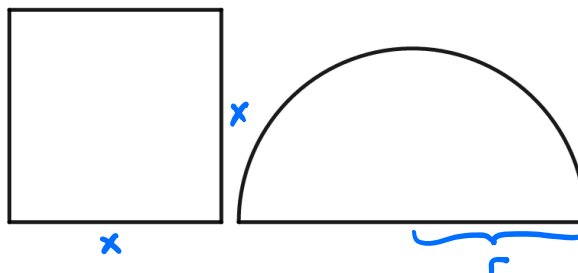
$$D'(x) = 2x - 2 = 0 \rightarrow x = 1$$



abs. min (only critical point)

$$D(1) = 1^2 - 2 \cdot 1 + 4 = \boxed{3}$$

- (b) A piece of wire 1 m long is cut into two pieces. One piece is bent into a square and the other is bent into a semicircle, as shown to the right. What length of wire should be used for the square so that the total area enclosed is a minimum. Be sure to use an appropriate test to show that the area you find is a minimum.

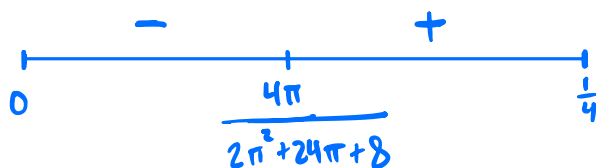


$$4x + 2r + \pi r = 1 \rightarrow r = \frac{1-4x}{2+\pi}$$

$$A(x) = x^2 + \frac{1}{2} \pi \left(\frac{1-4x}{2+\pi} \right)^2 \quad \text{on } \left[0, \frac{1}{4} \right]$$

$$A'(x) = 2x + \pi \left(\frac{1-4x}{2+\pi} \right) \cdot \frac{-4}{2+\pi} = 0 \rightarrow 2x + \frac{16\pi}{(2+\pi)^2} x = \frac{4\pi}{(2+\pi)^2}$$

$$\rightarrow x = \frac{\frac{4\pi}{(2+\pi)^2}}{2 + \frac{16\pi}{(2+\pi)^2}} = \frac{4\pi}{2\pi^2 + 24\pi + 8}$$



abs. min so

$$4x = \boxed{\frac{16\pi}{2\pi^2 + 24\pi + 8}}$$

B5 - Antiderivatives

Suppose an object's position function $s(t)$ satisfies $s(1) = 14$ and $s(e) = 2e$, and its acceleration as a function of time is given by the piecewise function

$$a(t) = \begin{cases} 12t^{-2} & 0 \leq t < 3 \\ \sin(t-3) & 3 \leq t \end{cases}$$

(a) Find an expression for $s(t)$ on the interval $[0, 3]$.

$$v(t) = -12 \cdot t^{-1} + C \quad \textcircled{2} - \textcircled{1}: -12 + C \cdot (e-1) = 2e - 14$$

$$s(t) = -12 \cdot \ln(t) + Ct + D$$

$$C = \frac{2e-2}{e-1} = 2$$

$$s(1) = -12 \cdot \ln(1) + C \cdot 1 + D = 14$$

$$\textcircled{1} C + D = 14$$

$$\rightarrow D = 14 - 2 = 12$$

$$s(e) = -12 \cdot \ln(e) + C \cdot e + D = 2e$$

$$\textcircled{2} -12 + C \cdot e + D = 2e$$

$$\boxed{s(t) = -12 \ln(t) + 2t + 12}$$

(b) Find an expression for the object's velocity function $v(t)$ on the interval $[3, \infty)$.

$$v(t) = \frac{-12}{t} + 2 \quad \text{on } [0, 3] \quad \text{so} \quad v(3) = \frac{-12}{3} + 2 = -2$$

$$v(t) = -\cos(t-3) + A$$

$$v(3) = -\cos(3-3) + A = -2$$

$$-1 + A = -2 \rightarrow A = -1$$

$$\boxed{v(t) = -\cos(t-3) - 1}$$

(c) Find the change in the object's position between $t = 3$ and $t = 4$.

$$s(t) = -\sin(t-3) - t + B$$

$$s(4) - s(3) = (-\sin(4-3) - 4 + \cancel{B}) - (-\sin(3-3) - 3 + \cancel{B})$$

$$\boxed{= -\sin(1) - 1}$$